AnnexureNo :

Practical: 2

Aim: Experiment of Packet Capture Tool: Wireshark.

WHAT IS WIRESHARK?

Wireshark is an open-source packet analyzer, which is used for education, analysis, software development, communication protocol development, and network troubleshooting.

It is used to track the packets so that each one is filtered to meet our specific needs. It is commonly called as a sniffer, network protocol analyzer, and network analyzer. It is also used by network security engineers to examine security problems.

Wireshark is a free to use application which is used to apprehend the data back and forth. It is often called as a free packet sniffer computer application. It puts the network card into an unselective mode, i.e., to accept all the packets which it receives.

USES OF WIRESHARK:

Wireshark can be used in the following ways:

1. It is used by network security engineers to examine security problems.
2. It allows the users to watch all the traffic being passed over the network.
3. It is used by network engineers to troubleshoot network issues.
4. It also helps to troubleshoot latency issues and malicious activities on your network.
5. It can also analyze dropped packets.
6. It helps us to know how all the devices like laptop, mobile phones, desktop, switch, routers, etc., communicate in a local network or the rest of the world.

HISTORY OF WIRESHARK:

In the late 1990's Gerald Combs, a computer science graduate of the University of Missouri-Kansas City was working for the small ISP (Internet Service Provider). The protocol at that time did not complete the primary requirements. So, he started writing ethereal and released the first version around 1998. The Network integration services owned the Ethernet trademark.

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FUNCTIONALITY OF WIRESHARK:

Wireshark is similar to TCP dump in networking. Tcpdump is a common packet analyzer which allows the user to display other packets and TCP/IP packets, being transmitted and received over a network attached to the computer. It has a graphic end and some sorting and filtering functions. Wireshark users can see all the traffic passing through the network. Wireshark can also monitor the unicast traffic which is not sent to the network's MAC address interface. But the switch does not pass all the traffic to the port. Hence, the promiscuous mode is not sufficient to see all the traffic. The various network taps or port mirroring is used to extend capture at any point. Port mirroring is a method to monitor network traffic. When it is enabled, the switch sends the copies of all the network packets present at one port to another port.

WHAT IS COLOR CODING IN WIRESHARK?

The packets in the Wireshark are highlighted with blue, black, and green colour These colours help users to identify the types of traffic. It is also called as packet colorization. The kinds of colouring rules in the Wireshark are temporary rules and permanent rules.

The temporary rules are there until the program is in active mode or until we quit the program. The permanent color rules are available until the Wireshark is in use or the next time you run the Wireshark. The steps to apply color filters will be discussed later in this topic.

FEATURES OF WIRESHARK:

- It is multi-platform software, i.e., it can run on Linux, Windows, OS X, FreeBSD, NetBSD, etc.
- It is a standard three-pane packet browser.
- It performs deep inspection of the hundreds of protocols.
- It often involves live analysis, i.e., from the different types of the network like the Ethernet, loopback, etc., we can read live data.
- It has sort and filter options which makes ease to the user to view the data. It is also useful in VoIP analysis.
- It can also capture raw USB traffic.

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Various settings, like timers and filters, can be used to filter the output.

- It can only capture packet on the PCAP (an application programming interface used to capture the network) supported networks.
- Wireshark supports a variety of well-documented capture file formats such as the PcapNg and Libpcap. These formats are used for storing the captured data.
- It is the no.1 piece of software for its purpose. It has countless applications ranging from the tracing down, unauthorized traffic, firewall settings, etc.

INSTALLATION OF WIRESHARK SOFTWARE:

Below are the steps to install the Wireshark software on the computer: Open the web browser. Search for 'Download Wireshark.'

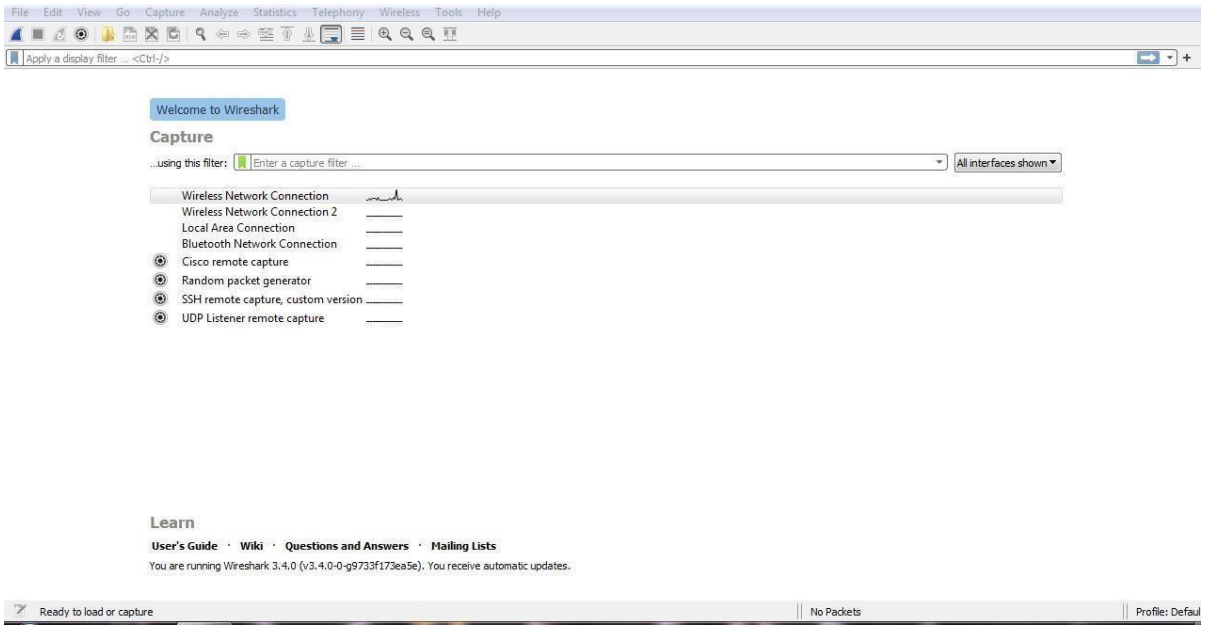
Select the Windows installer according to your system configuration, either 32-bt or 64-bit. Save the program and close the browser.

Now, open the software, and follow the install instruction by accepting the license. The Wireshark is ready for use.

On the network and Internet settings option, we can check the interface connected to our computer.

If you are Linux users, then you will find Wireshark in its package repositories.

By selecting the current interface, we can get the traffic traversing through that interface. The version used here is 3.0.3. This version will open as:



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The screen/interface of the Wireshark is divided into five parts:

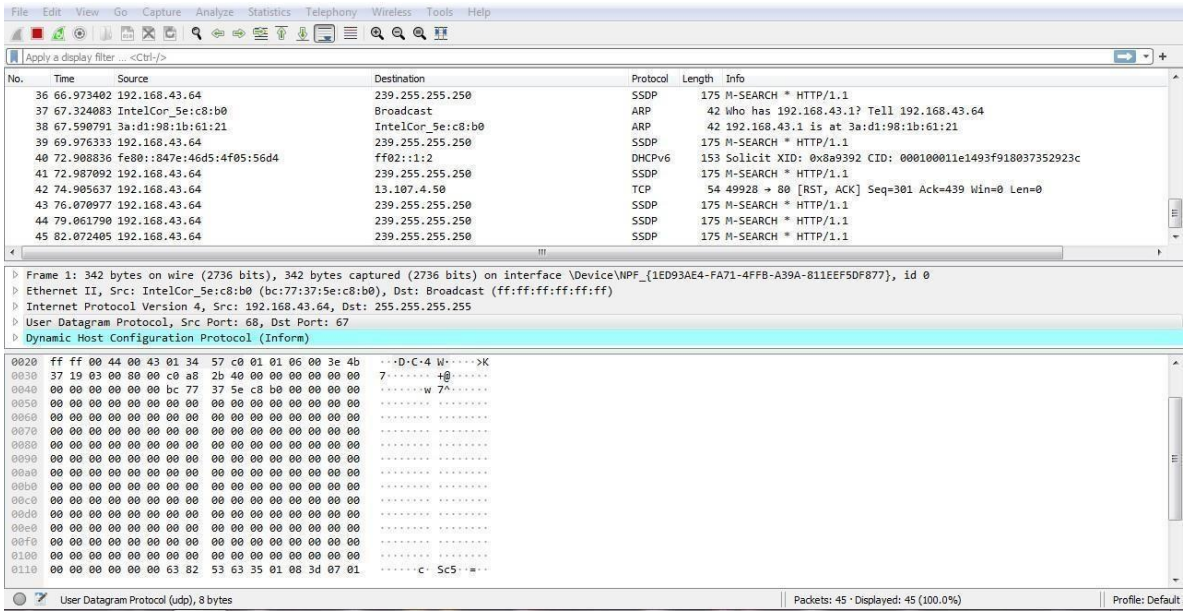
First part contains a menu bar and the options displayed below it. This part is at the top of the window. File and the capture menus options are commonly used in

Wireshark. The capture menu allows to start the capturing process. And the File menu is used to open and save a capture file.

The second part is the packet listing window. It determines the packet flow or the captured packets in the traffic. It includes the packet number, time, source, destination, protocol, length, and info. We can sort the packet list by clicking on the column name. Next comes the packet header- detailed window. It contains detailed information about the components of the packets. The protocol info can also be expanded or minimized according to the information required.

The bottom window called the packet contents window, which displays the content in ASCII and hexadecimal format.

At last, is the filter field which is at the top of the display. The captured packets on the screen can be filtered based on any component according to your requirements. For example, if we want to see only the packets with the HTTP protocol, we can apply filters to that option. All the packets with HTTP as the protocol will only be displayed on the screen, shown below:



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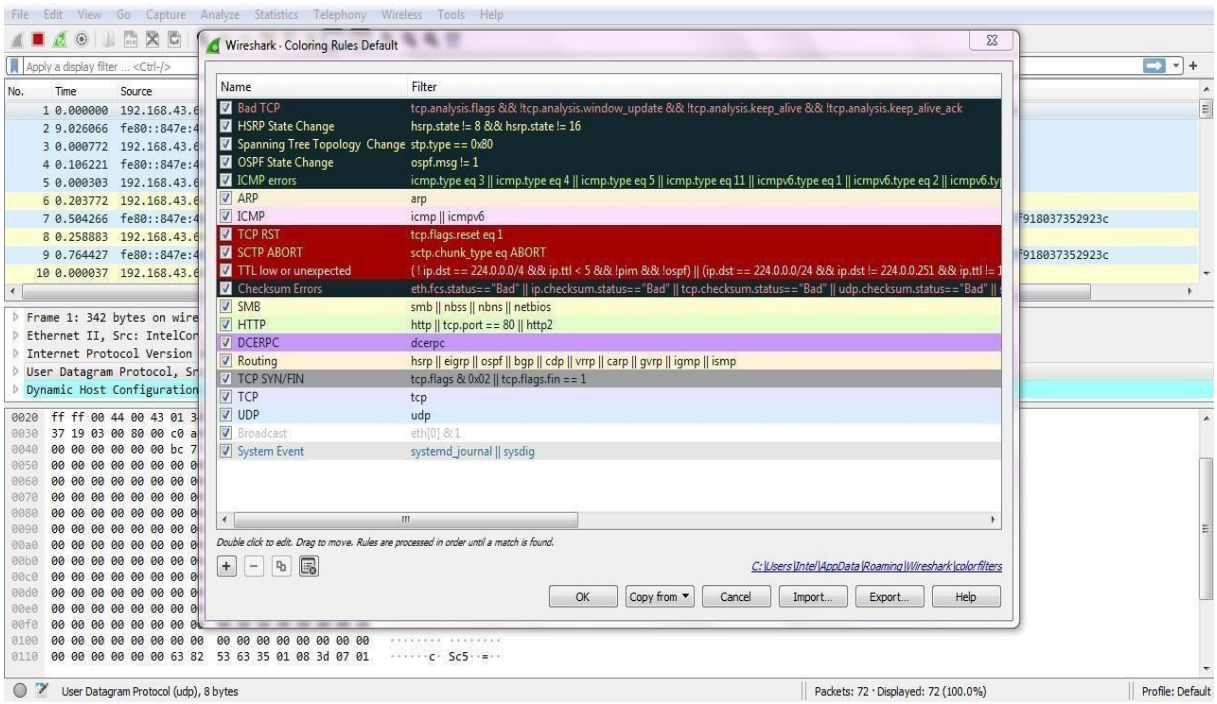
If you want to filter according to the source, right-click on the source you want to filter and select 'Apply as Filter' and choose '...and filter.' Steps for the permanent colorization are: click on the 'View' option on the menu bar and select 'Coloring Rules.' The table will appear like the image shown below:



For the network administrator job, advanced knowledge of Wireshark is considered as the requirements. So, it is essential to understand the concepts of the software. It contains these 20 default coloring rules which can be added or removed according to the requirements. Select the option 'View' and then choose 'Colorize Packet List,' which is used to toggle the color on and off.

MOST USED FILTER IN WIRESHARK:

Whenever we type any commands in the filter command box, it turns green if your command is correct. It turns red if it is incorrect or the Wireshark does not recognize your command.

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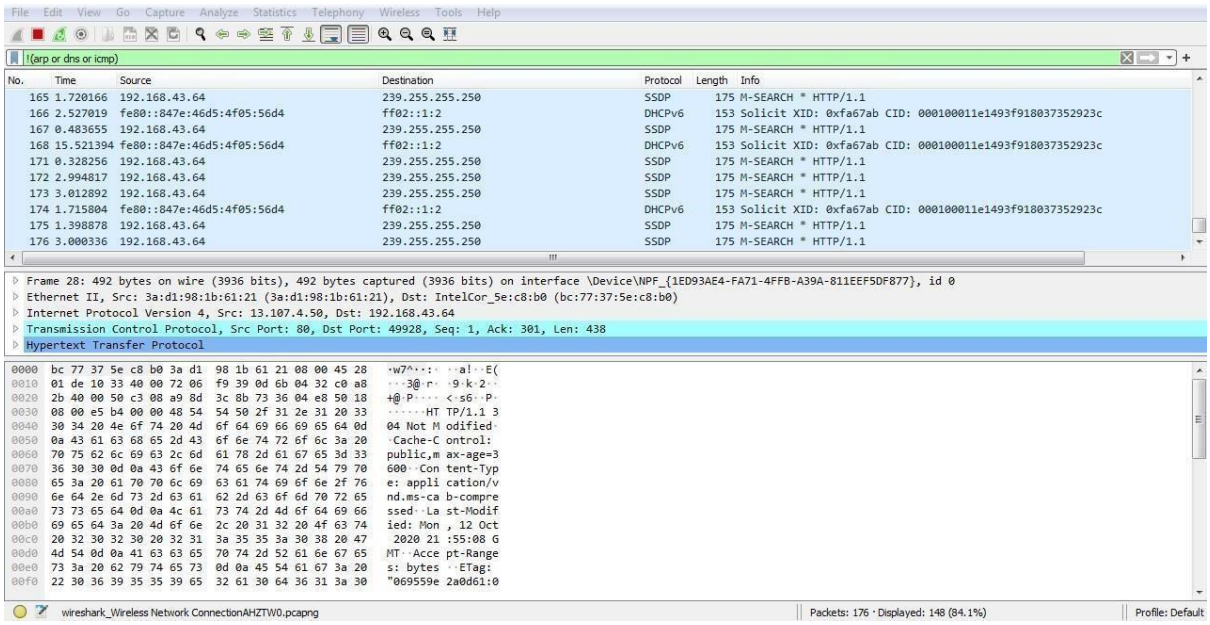
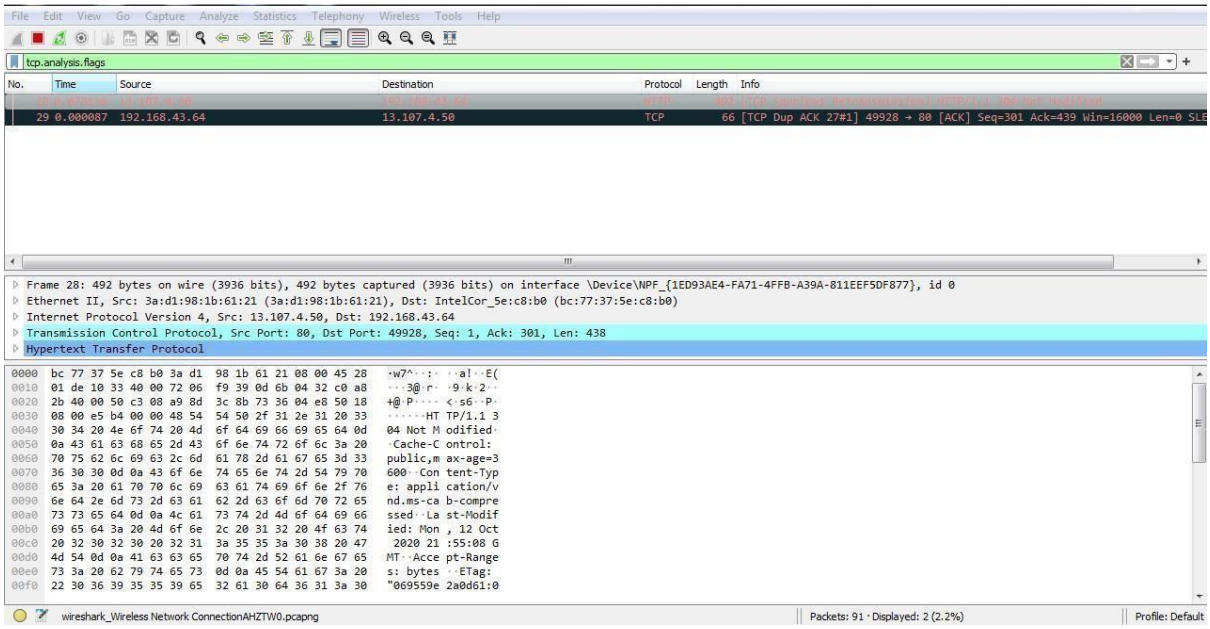

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
!(arp or dns or icmp)
No. Time Source Destination Protocol Length Info
165 1.720166 192.168.43.64 239.255.255.250 SSDP 175 M-SEARCH * HTTP/1.1
166 2.527019 fe80::847e:46d5:4f05:56d4 ff02::1:2 DHCPv6 153 Solicit XID: 0xfa67ab CID: 000100011e1493f918037352923c
167 0.483655 192.168.43.64 239.255.255.250 SSDP 175 M-SEARCH * HTTP/1.1
168 15.521394 fe80::847e:46d5:4f05:56d4 ff02::1:2 DHCPv6 153 Solicit XID: 0xfa67ab CID: 000100011e1493f918037352923c
171 0.328256 192.168.43.64 239.255.255.250 SSDP 175 M-SEARCH * HTTP/1.1
172 2.994817 192.168.43.64 239.255.255.250 SSDP 175 M-SEARCH * HTTP/1.1
173 3.012892 192.168.43.64 239.255.255.250 SSDP 175 M-SEARCH * HTTP/1.1
174 1.715804 fe80::847e:46d5:4f05:56d4 ff02::1:2 DHCPv6 153 Solicit XID: 0xfa67ab CID: 000100011e1493f918037352923c
175 1.398878 192.168.43.64 239.255.255.250 SSDP 175 M-SEARCH * HTTP/1.1
176 3.000336 192.168.43.64 239.255.255.250 SSDP 175 M-SEARCH * HTTP/1.1
Frame 28: 492 bytes on wire (3936 bits), 492 bytes captured (3936 bits) on interface \Device\NPF_{1ED93AE4-FA71-4FFB-A39A-811EEF5DF877}, id 0
Ethernet II, Src: 3a:d1:98:1b:61:21 (3a:d1:98:1b:61:21), Dst: IntelCor_5e:c8:b0 (bc:77:37:5e:c8:b0)
Internet Protocol Version 4, Src: 13.107.4.50, Dst: 192.168.43.64
Transmission Control Protocol, Src Port: 80, Dst Port: 49928, Seq: 1, Ack: 301, Len: 438
Hypertext Transfer Protocol
0000 bc 77 37 5e c8 b0 3a d1 98 1b 61 21 08 00 45 28 ·w7^··:· ··a!··E(
0010 01 de 10 33 40 00 72 06 f9 39 0d 6b 04 32 c0 a8 ···3@·r· ·9·k·2··
0020 2b 40 00 50 c3 08 a9 8d 3c 8b 73 36 04 e8 50 18 +@·P···· <·s6··P·
0030 08 00 e5 b4 00 00 48 54 54 50 2f 31 2e 31 20 33 ····· HT TP/1.1 3
0040 30 34 20 4e 6f 74 20 4d 6f 64 69 66 69 65 64 0d 04 Not M odified·
0050 0a 43 61 63 68 65 2d 43 6f 6e 74 72 6f 6c 3a 20 ·Cache-C ontrol:
0060 70 75 62 6c 69 63 2c 6d 61 78 2d 61 67 65 3d 33 public,m ax-age=3
0070 36 30 30 0d 0a 43 6f 6e 74 65 6e 74 2d 54 79 70 600··Con tent-Typ
0080 65 3a 20 61 70 70 6c 69 63 61 74 69 6f 6e 2f 76 e: appli cation/v
0090 6e 64 2e 6d 73 2d 63 61 62 2d 63 6f 6d 70 72 65 nd.ms-ca b-compre
00a0 73 73 65 64 0d 0a 4c 61 73 74 2d 4d 6f 64 69 66 ssed··La st-Modif
00b0 69 65 64 3a 20 4d 6f 6e 2c 20 31 32 20 4f 63 74 ied: Mon , 12 Oct
00c0 20 32 30 32 30 20 32 31 3a 35 35 3a 30 38 20 47 2020 21 :55:08 G
00d0 4d 54 0d 0a 41 63 63 65 70 74 2d 52 61 6e 67 65 MT··Acce pt-Range
00e0 73 3a 20 62 79 74 65 73 0d 0a 45 54 61 67 3a 20 s: bytes ··ETag:
00f0 22 30 36 39 35 35 39 65 32 61 30 64 36 31 3a 30 "069559e 2a0d61:0
wireshark_Wireless Network ConnectionAHZTW0.pcapng
Packets: 176 · Displayed: 148 (84.1%)
Profile: Default


File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.analysis.flags
No. Time Source Destination Protocol Length Info
29 0.000087 192.168.43.64 13.107.4.50 TCP 66 [TCP Dup ACK 27#1] 49928 → 80 [ACK] Seq=301 Ack=439 Win=16000 Len=0 SLE
Frame 28: 492 bytes on wire (3936 bits), 492 bytes captured (3936 bits) on interface \Device\NPF_{1ED93AE4-FA71-4FFB-A39A-811EEF5DF877}, id 0
Ethernet II, Src: 3a:d1:98:1b:61:21 (3a:d1:98:1b:61:21), Dst: IntelCor_5e:c8:b0 (bc:77:37:5e:c8:b0)
Internet Protocol Version 4, Src: 13.107.4.50, Dst: 192.168.43.64
Transmission Control Protocol, Src Port: 80, Dst Port: 49928, Seq: 1, Ack: 301, Len: 438
Hypertext Transfer Protocol
0000 bc 77 37 5e c8 b0 3a d1 98 1b 61 21 08 00 45 28 ·w7^··:· ··a!··E(
0010 01 de 10 33 40 00 72 06 f9 39 0d 6b 04 32 c0 a8 ···3@·r· ·9·k·2··
0020 2b 40 00 50 c3 08 a9 8d 3c 8b 73 36 04 e8 50 18 +@·P···· <·s6··P·
0030 08 00 e5 b4 00 00 48 54 54 50 2f 31 2e 31 20 33 ····· HT TP/1.1 3
0040 30 34 20 4e 6f 74 20 4d 6f 64 69 66 69 65 64 0d 04 Not M odified·
0050 0a 43 61 63 68 65 2d 43 6f 6e 74 72 6f 6c 3a 20 ·Cache-C ontrol:
0060 70 75 62 6c 69 63 2c 6d 61 78 2d 61 67 65 3d 33 public,m ax-age=3
0070 36 30 30 0d 0a 43 6f 6e 74 65 6e 74 2d 54 79 70 600··Con tent-Typ
0080 65 3a 20 61 70 70 6c 69 63 61 74 69 6f 6e 2f 76 e: appli cation/v
0090 6e 64 2e 6d 73 2d 63 61 62 2d 63 6f 6d 70 72 65 nd.ms-ca b-compre
00a0 73 73 65 64 0d 0a 4c 61 73 74 2d 4d 6f 64 69 66 ssed··La st-Modif
00b0 69 65 64 3a 20 4d 6f 6e 2c 20 31 32 20 4f 63 74 ied: Mon , 12 Oct
00c0 20 32 30 32 30 20 32 31 3a 35 35 3a 30 38 20 47 2020 21 :55:08 G
00d0 4d 54 0d 0a 41 63 63 65 70 74 2d 52 61 6e 67 65 MT··Acce pt-Range
00e0 73 3a 20 62 79 74 65 73 0d 0a 45 54 61 67 3a 20 s: bytes ··ETag:
00f0 22 30 36 39 35 35 39 65 32 61 30 64 36 31 3a 30 "069559e 2a0d61:0
wireshark_Wireless Network ConnectionAHZTW0.pcapng
Packets: 91 · Displayed: 2 (2.2%)
Profile: Default

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Practical: 3

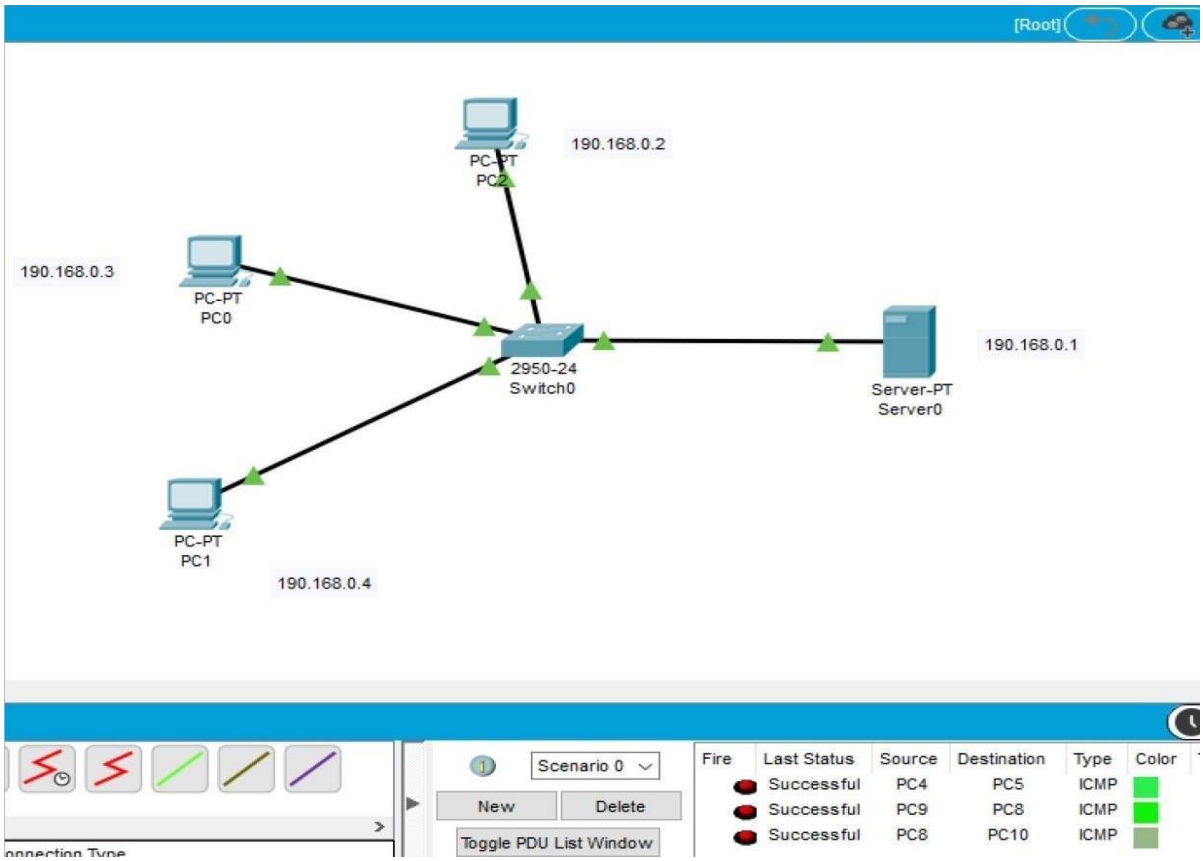
Aim: To study behaviour of generic devices used for networking: (cisco packet tracer).

THEORY:

In this Practical we are going to learn the generic devices of Computer Networking.

1. WEB SERVER:

A server is a computer connected to a network of other workstations called 'clients'. Client computers request information from the server over the network. Servers tend to have more storage, memory and processing power than a normal workstation.



USES:

A server is a computer connected to a network of other workstations called ‘clients. Client computers request information from the server over the network. Servers tend to have more

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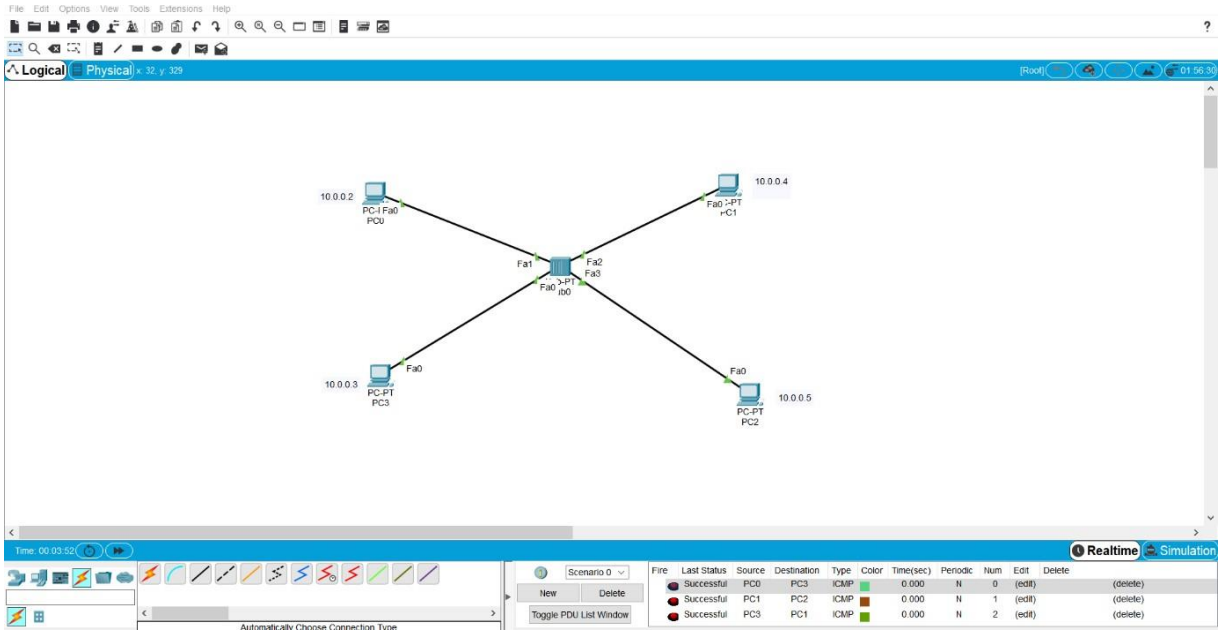
storage, memory and processing power than a normal workstation.

2. HUB:

A hub is a physical layer networking device which is used to connect multiple devices in a network. They are generally used to connect computers in a LAN. A hub has many ports in it. A computer which intends to be connected to the network is plugged in to one of these ports.



USES:

A hub is a physical layer networking device which is used to connect multiple devices in a network. They are generally used to connect computers in a LAN. A computer which intends to be connected to the network is plugged in to one of these ports.

3. ROUTERS:

A router receives and sends data on computer networks. Routers are sometimes confused with network hubs, modems, or network switches. However, routers can combine the functions of these components, and connect with these devices, to improve Internet access or help create business networks.

USES:

AnnexureNo :

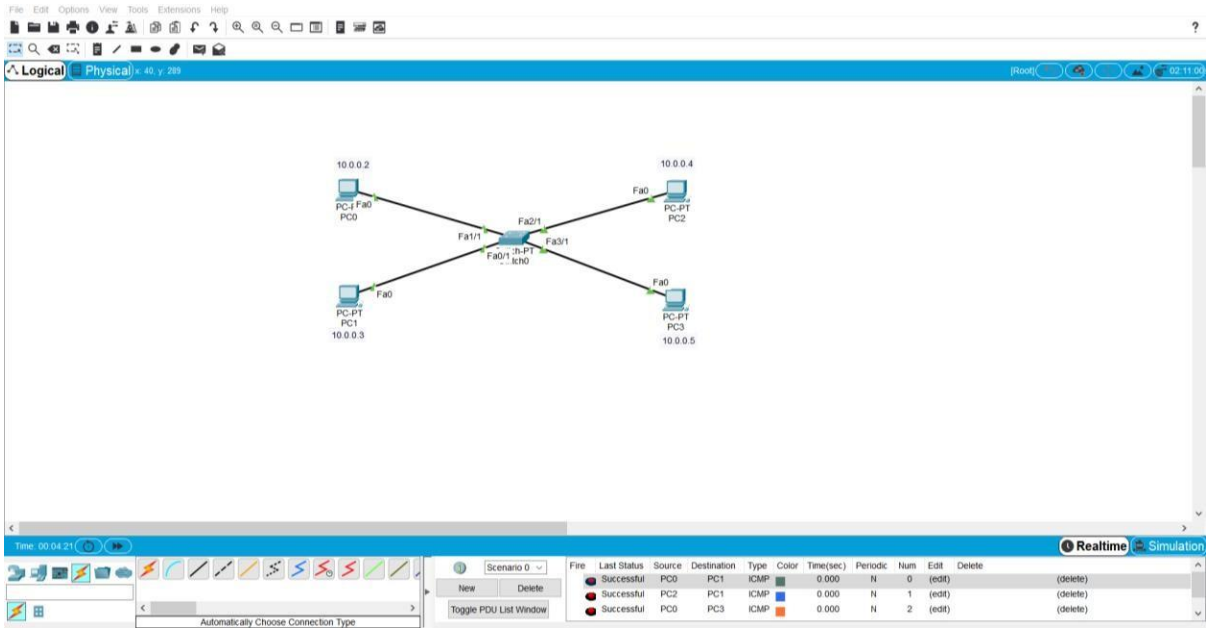
A router receives and sends data on computer networks. Routers are sometimes confused with network hubs, modems, or network switches. However, routers can combine the functions of these components, and connect with these devices, to improve Internet access or help create business networks.

4. SWITCHES:

A switch is a device in a computer network that connects other devices together. Multiple data cables are plugged into a switch to enable communication between different networked devices. Switches may also operate at higher layers of the OSI model, including the network layer and above.

USES:

Switches are key building blocks for any network. They connect multiple devices, such as computers, wireless access points, printers, and servers; on the same network within a building or campus. A switch enables connected devices to share information and talk to each other.



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5. BRIDGE:

A network bridge is a computer networking device that creates a single aggregate network from multiple communication networks or network segments Routing allows multiple networks to communicate independently and yet remain separate, whereas bridging connects two separate networks as if they were a single network.

USES:

A network bridge is a computer networking device that creates a single aggregate network from multiple communication networks or network segments. Routing allows multiple networks to communicate independently and yet remain separate, whereas bridging connects two separate networks as if they were a single network.

6. SERVER:

A server is a computer or system that provides resources, data, services, or programs to other computers, known as clients, over a network. In theory, whenever computers share resources with client machines, they are considered servers.

7. WIRELESS DEVICES:

Devices such as computers, tablets, and phones are common Clients on a network. Access Points (Master) most wireless networks are made using Access Points – devices that host and control the wireless connection for laptops, tablets, or smart phones.

8. LAPTOP:

A laptop computer, sometimes called a notebook computer by manufacturers, is a battery- or AC powered personal computer generally smaller than a briefcase that can easily be transported and conveniently used in temporary spaces such as on airplanes, in libraries, temporary offices, and at meetings.

9. IP PHONE:

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IP telephony refers to any phone system that uses an internet connection to send and receive voice data. Unlike a regular telephone that uses landlines to transmit analog signals, IP phones connect to the internet via a router and modem.

10. Modem:

Modem is a device or program that enables a computer to transmit data over telephone or cable lines. Computer information is stored digitally, whereas information transmitted over telephone lines is transmitted in the form of analogy waves.

11. Host:

Host is a computer or other devices connected to a network. A host may work as a server offering information resources, services and application to users or other hosts.

12. Repeater:

A repeater operates at the physical layer. Its job is to regenerate the signal over the same network before the signal becomes too weak or corrupted so as to extend the length to which the signal can be transmitted over the same network.

13. Automatically chose connection:

lead to automatic connect and disconnect actions on Wi-Fi, in the Windows user interface but can be configured by using the at least one active Internet connection to a preferred type of network.

14. Console:

The console is a connection over serial port connection that allows a person access to a computer or network device console. Usually, a console is accessed over an SSH connection.

15. Copper straight-through:

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Straight-through cable is a type of twisted pair copper wire cable for local area network (LAN) use for which the RJ-45 connectors at each end have the same pinout (i.e., arrangement of conductors).

16. Copper Crossover:

A crossover cable is a type of twisted pair copper wire cable for LANs (local area network) in which the wires on the cable are crossed over so that the receive signal pins on the RJ-45 connector on one end are connected to the transmit signal pins on the RJ-45 connector on the other end.

17. Fiber cable:

Fiber cables transmit pulses of light instead of electrical signals, so the terminations must be much more precise. Instead of merely allowing pins to make metal-to-metal contact, fiber optic connectors must align microscopic glass fiber perfectly in order to allow for communication.

18. Phone:

Phone a standard specifies both the physical connector and how it is wired, when modulator connectors are used, the latch release of the connector should be on the ridge side of flat phone wire in order to maintain polarity.

19. Coaxial cable:

Coaxial cable is **used as** a transmission line for radio frequency signals, taking a round cross section of the cable, one would find a signal centre solid wire symmetrically surrounded by a foil conductor.

20. Serial DCE data:

circuit training, data communications or data carrier equipment this is a modem or more generally adapter. It has to transmit the clock signal which controls the data rate.

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21. DTE

DTE stands for data terminal equipment which is generally a terminal or a computer. It is a device that acts as an information source or an information sink for the binary digital data.

21. Octal:

Octal is a term that describes a base 8 number system. It consists of eight single digit numbers:0,1,2,3,4,5,6 and 7. It can connect and access server to several different Cisco devices so that it can configure without plugging and unplugging.

22. USB:

USB is an industry standard that establishes specifications for cables and connectors and protocols for connection, communication and power supply between computers, peripheral devices and other computers.

PC to PC Connection:

Steps:

- ✓ Drag 2 PC's from end components.
- ✓ Assign a unique IP address to each PC.
- ✓ Drag a switch and connect 2 PC with a switch using copper straight through cable by fast ethernet ports.

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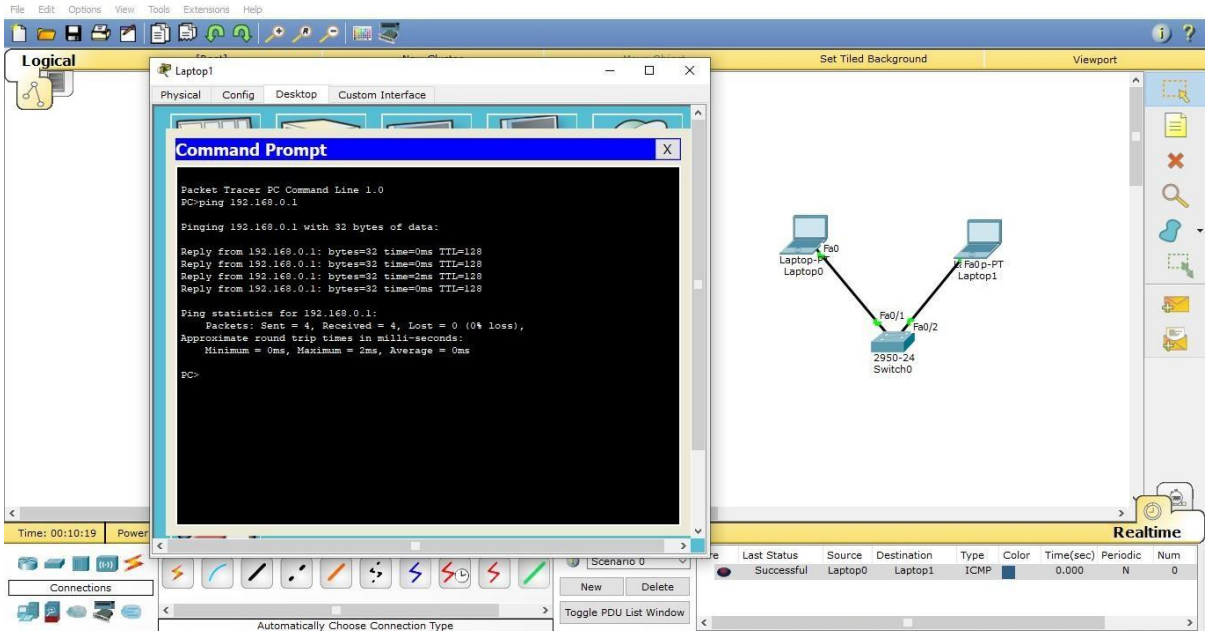
Logical
Laptop1
Physical
Config
Desktop
Custom Interface
Command Prompt
Packet Tracer PC Command Line 1.0
PC>ping 192.168.0.1
Pinging 192.168.0.1 with 32 bytes of data:
Reply from 192.168.0.1: bytes=32 time=0ms TTL=128
Reply from 192.168.0.1: bytes=32 time=0ms TTL=128
Reply from 192.168.0.1: bytes=32 time=2ms TTL=128
Reply from 192.168.0.1: bytes=32 time=0ms TTL=128
Ping statistics for 192.168.0.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 2ms, Average = 0ms
PC>
Laptop-PT
Laptop0
Fa0
Laptop-PT
Laptop1
Fa0/1
Fa0/2
2950-24
Switch0
Set Tiled Background
Viewport
Time: 00:10:19
Realtime
Connections
Automatically Choose Connection Type
New
Delete
Toggle PDU List Window
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Successful
Laptop0
Laptop1
ICMP
0.000
N
0